Computer Vision

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Introduction



Introduction

Why are we here?

We live in a world full of images and videos



Computers need to see and understand



Human: Eyes + Brain >> Understanding



Computers: we want to understand how they see"!



Computer Vision



Goals

What You'll learn

- Basics
- Applications
- Roadmap
- Tools and Resources

Let's Dive In!

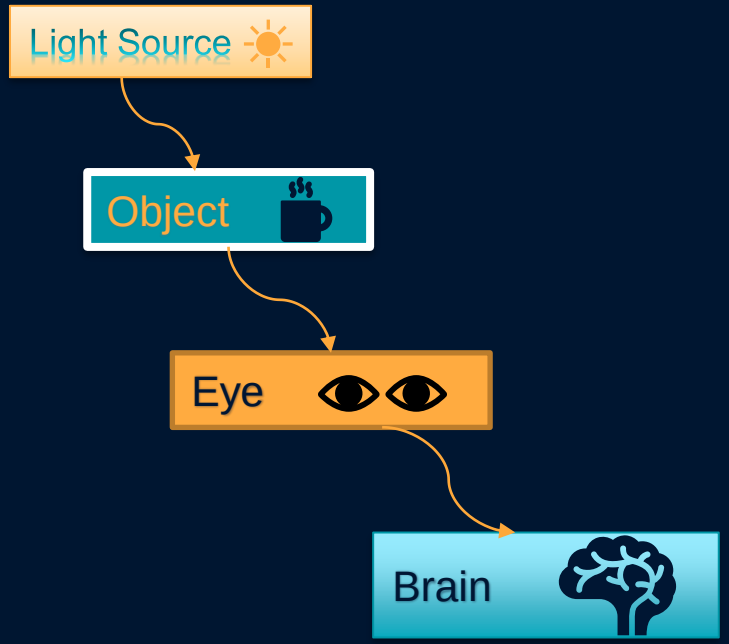


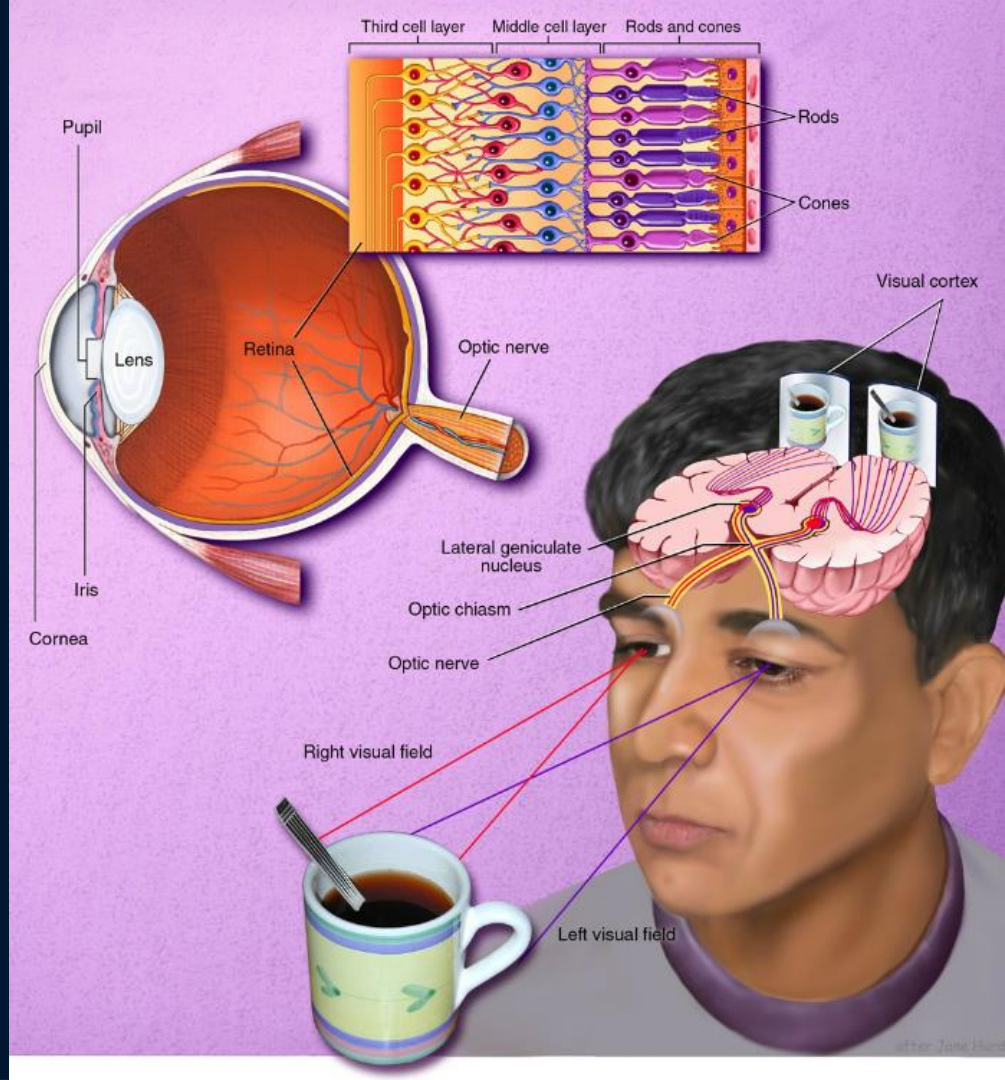
Basics



Human Vision

How we see?

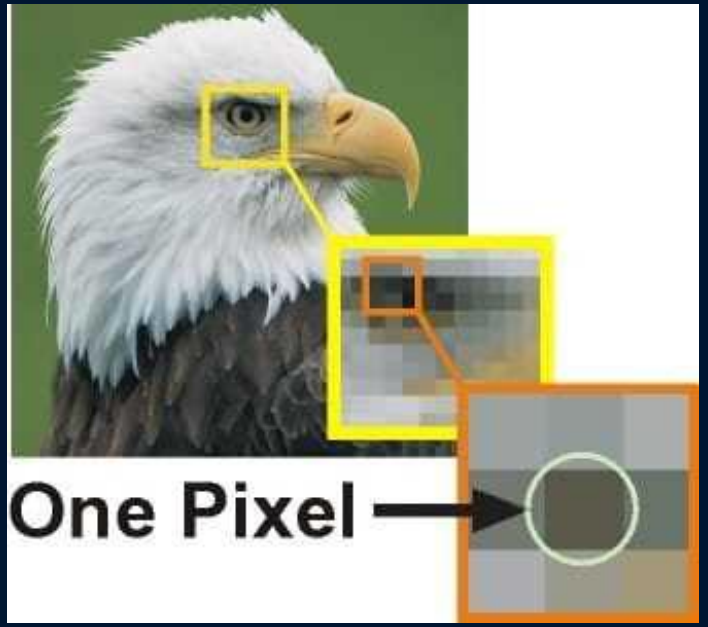


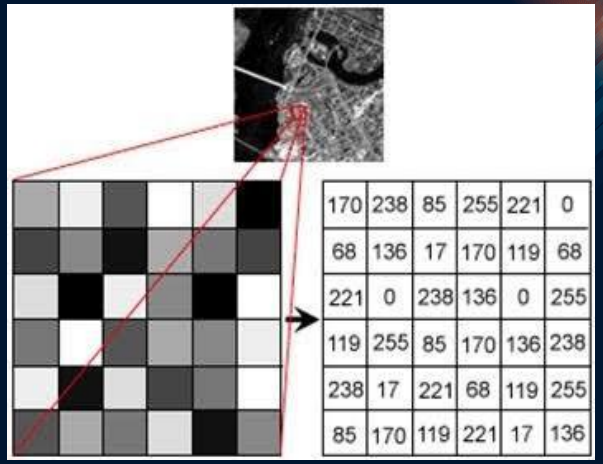


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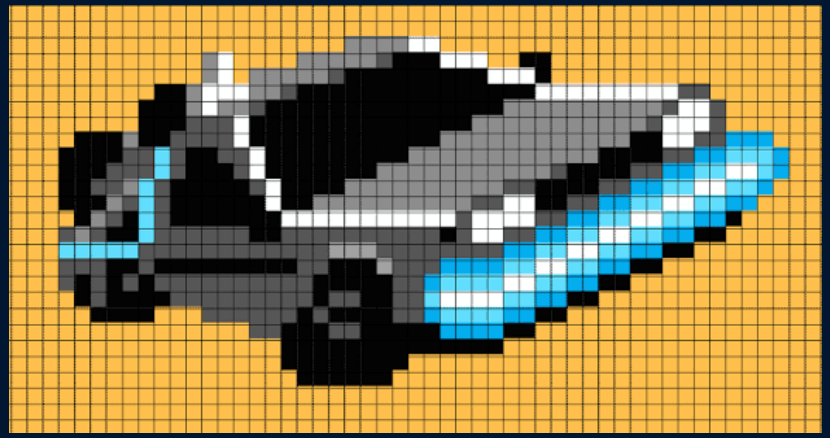
Digital Images

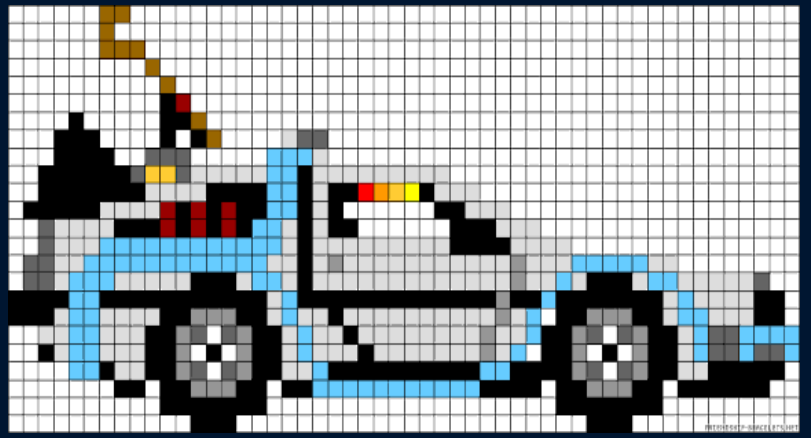
- Pixel
- Color
- Intensity
- Matrix





Digital Images

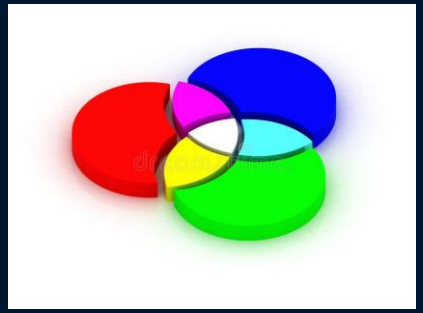


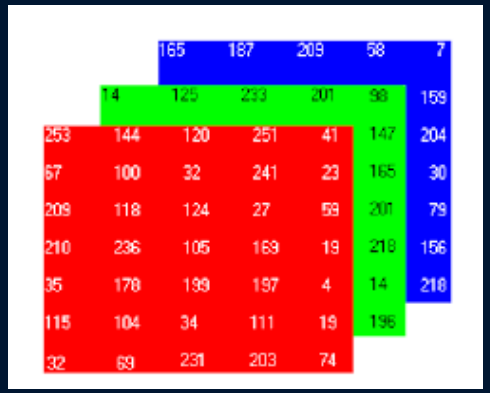


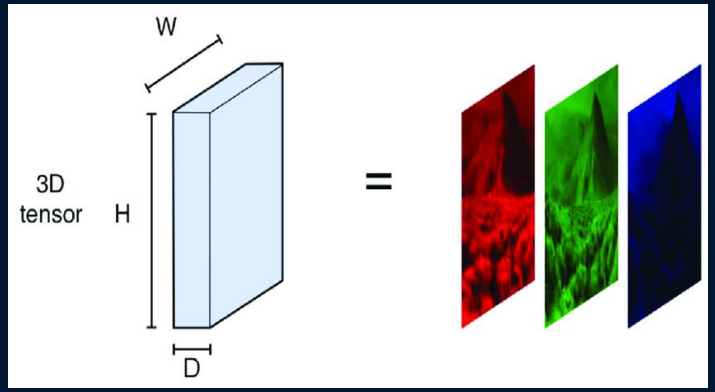
Color Images

RGB

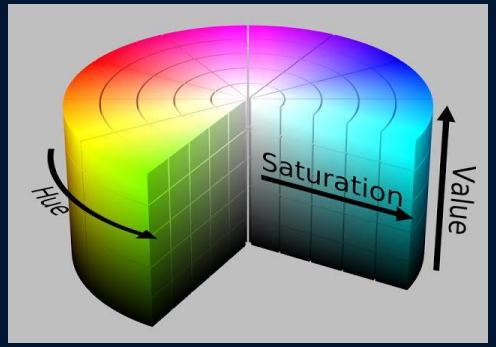
- R: Red
- G: Green
- B: Blue

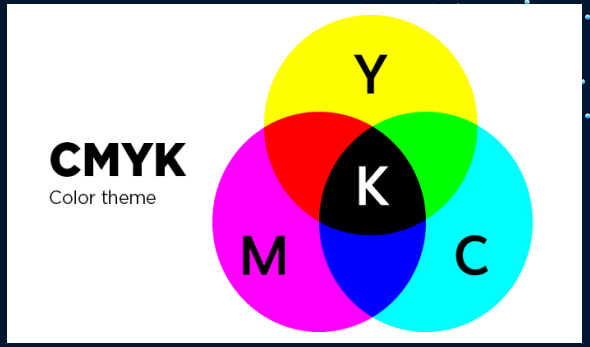


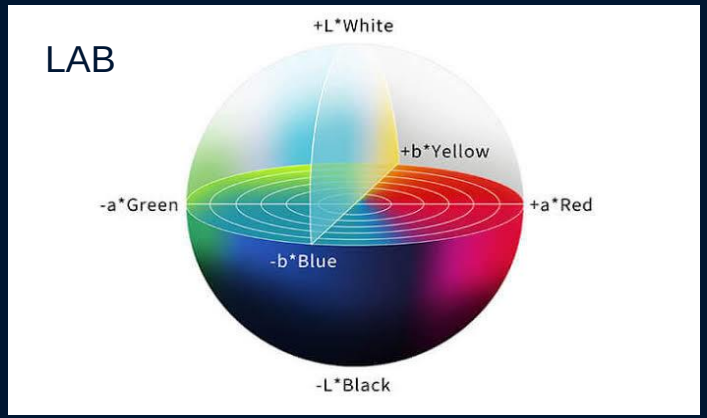


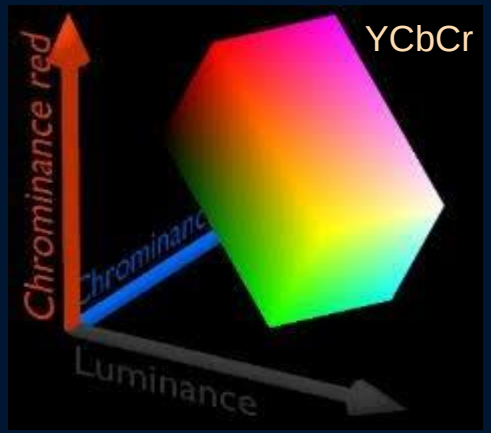


Other Color Model

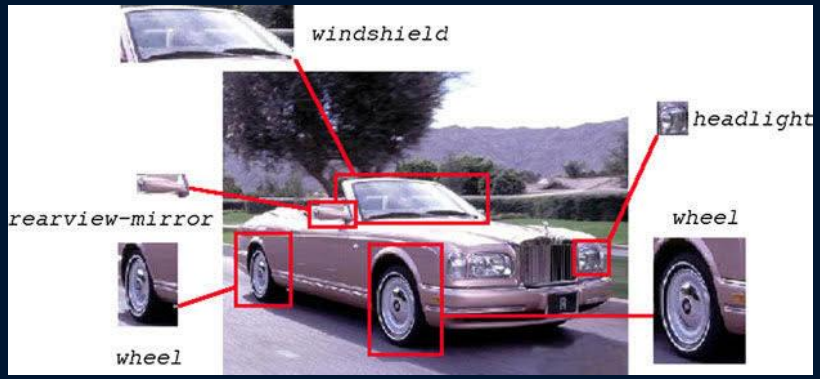


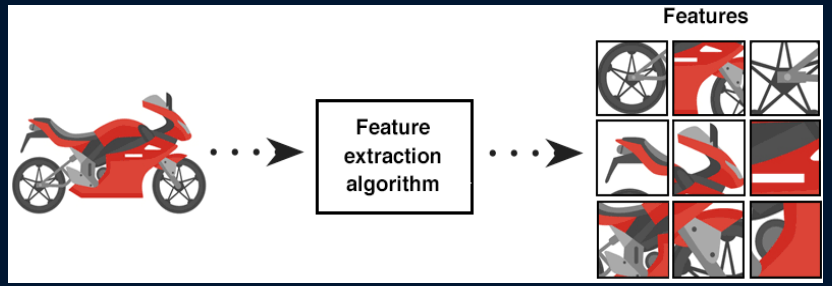






Human Vision

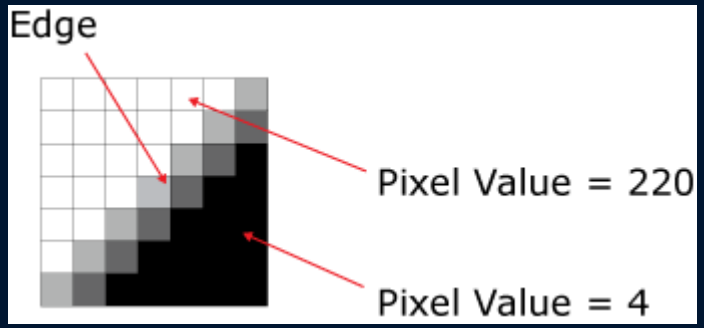






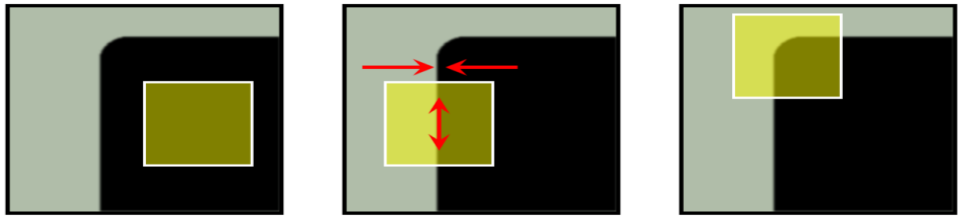


Features





Corner Detection: Basic idea



"flat" region: no change in all directions

"edge" : no change along the edge direction

"corner" : significant change in all directions with small shift

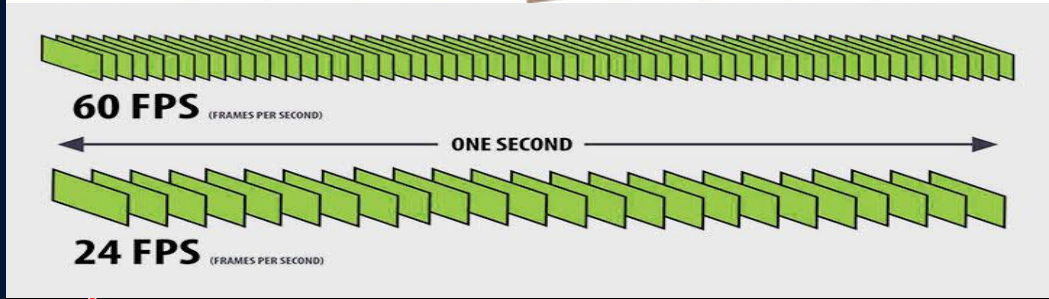
Video Basics

What is video?

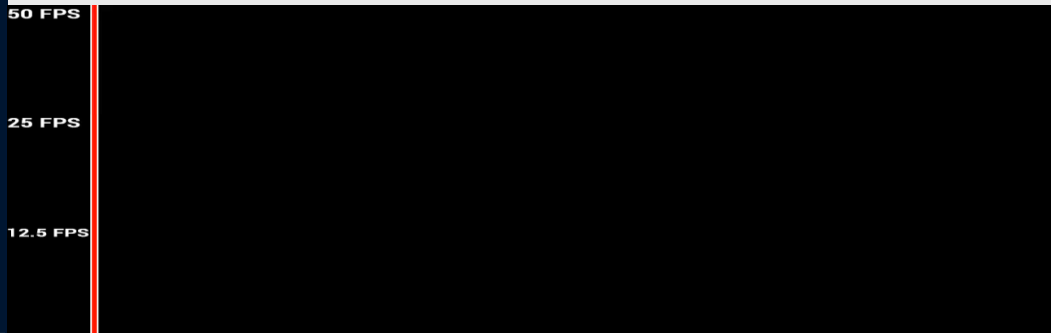
Video = Frame + Time







Frame Rate:
Frame Per Second(FPS)



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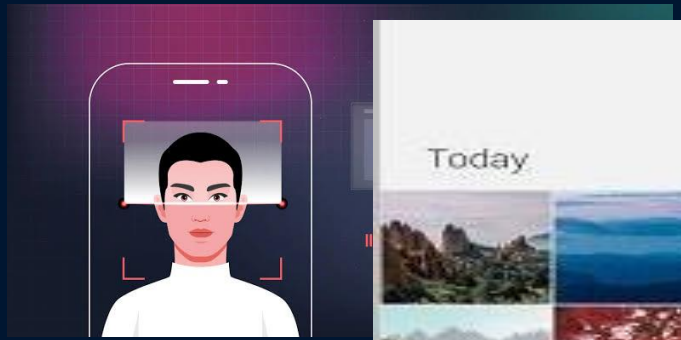
Application

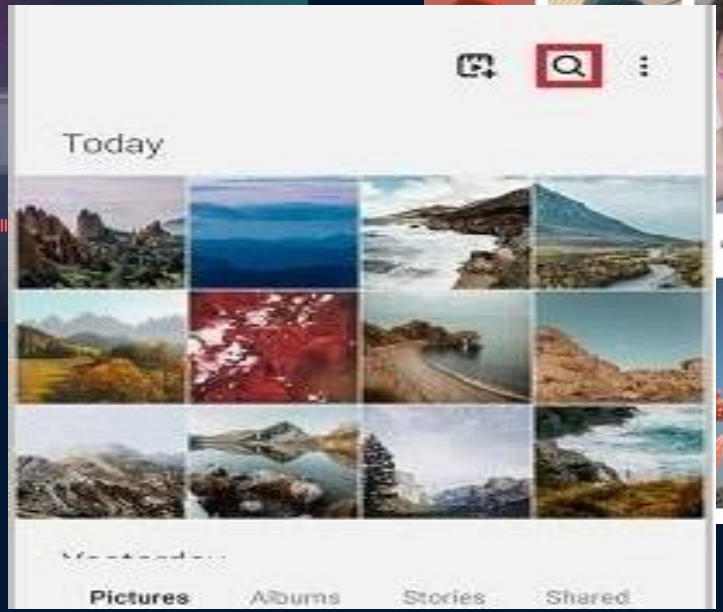


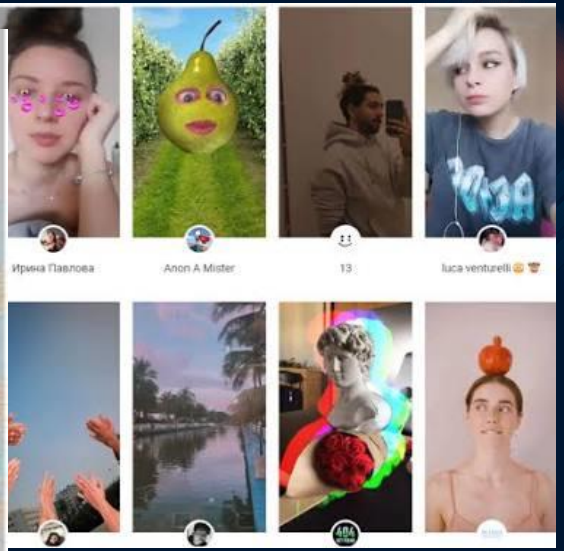
Application

Daily Life

- Unlocking phone with Face Recognition
- Instagram and Snapchat image filters
- Categorizing and searching photos gallery







Application

Transportation

- Self-driving cars like Tesla
- Driver assistance systems
- Traffic control cameras
- License plate reading



Transportation

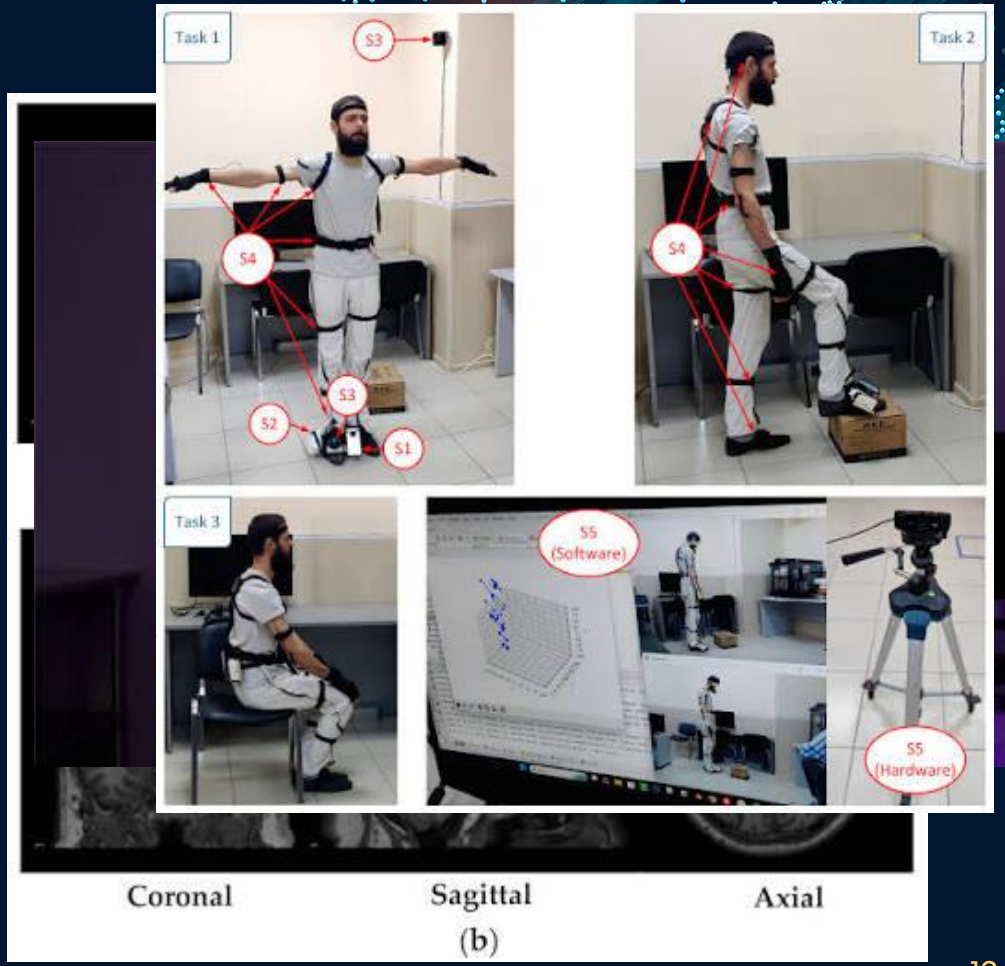
Self driving cars



Application

Medicine and Health

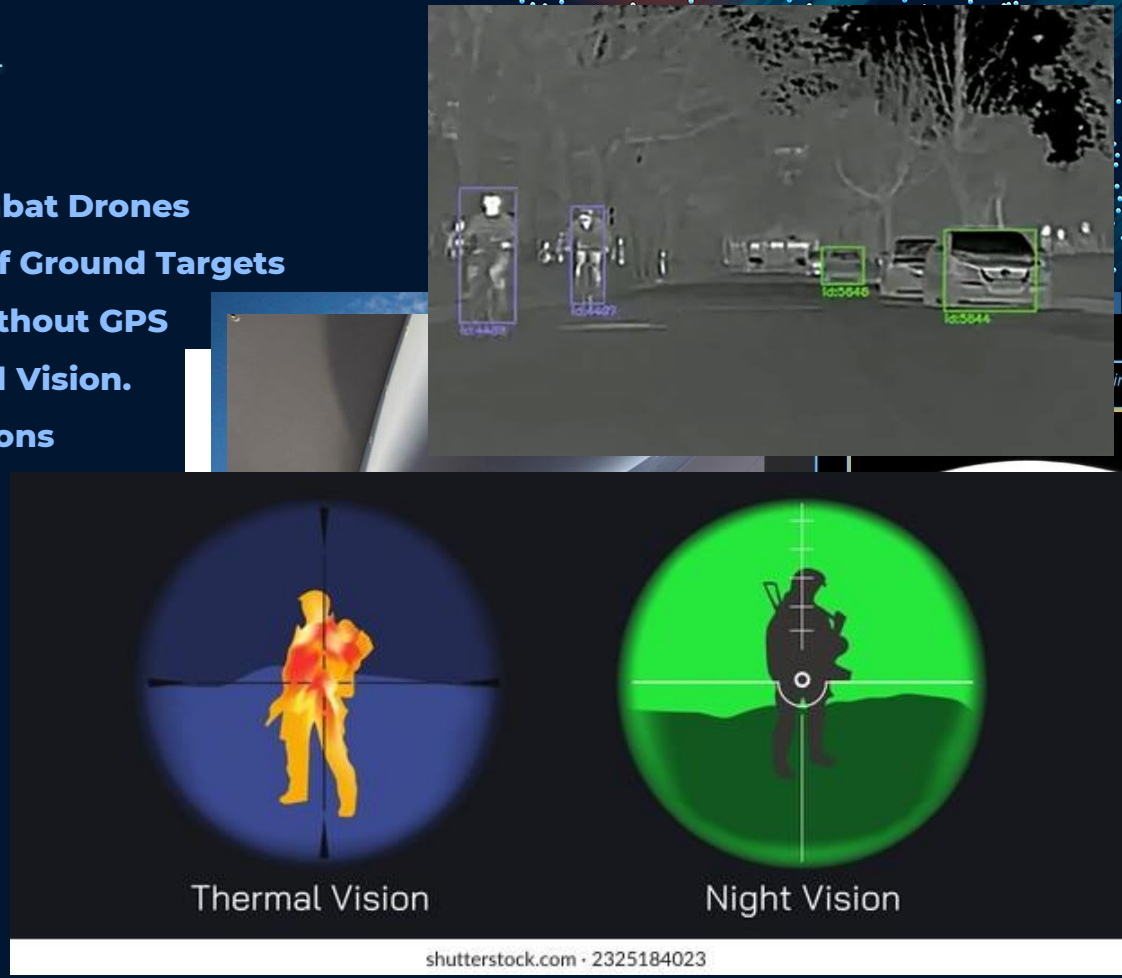
- Diagnosis of diseases from medical images (MRI, X-ray)
- Robotic surgery with precise computer vision
- Monitoring of patient movements (e.g. for physiotherapy)



Application

Military

- Reconnaissance and Combat Drones
- Detection and Tracking of Ground Targets
- Automatic Navigation Without GPS
- Night Vision with Infrared Vision.
- Smart Missiles and Weapons



Military

Changing warfare, small drones



Application

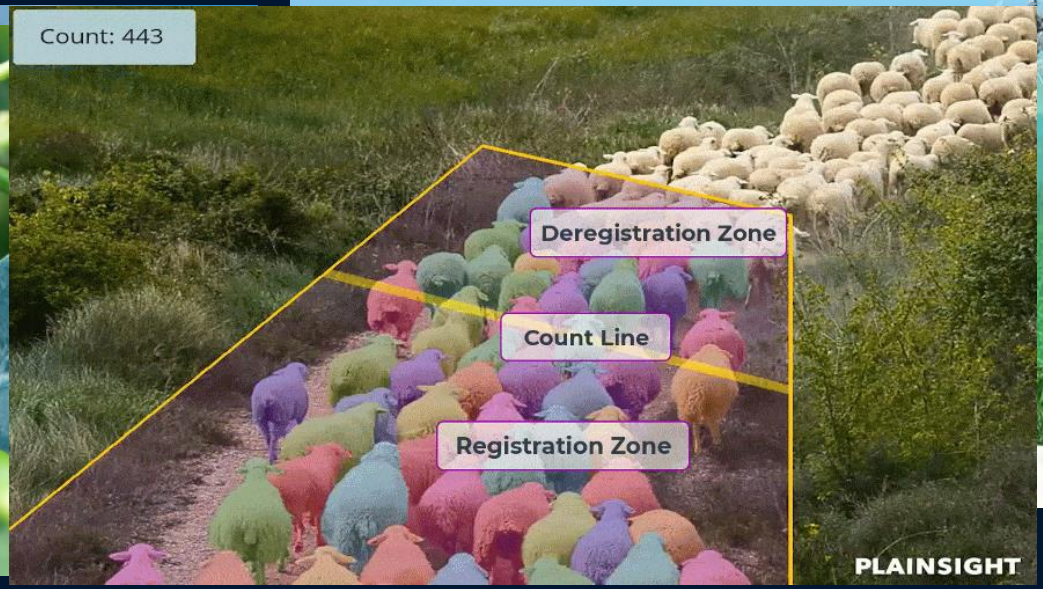
Agriculture and Environment

- Plant Health Detection with Drones
- Pest and Weed Identification
- Counting Livestock or Animals in the Wild

Start-ups specializing in drone technology are **Encouraging Farmers** to use drones for **Monitoring Crops** and **Spraying Fertilizers** and **Pesticides**.



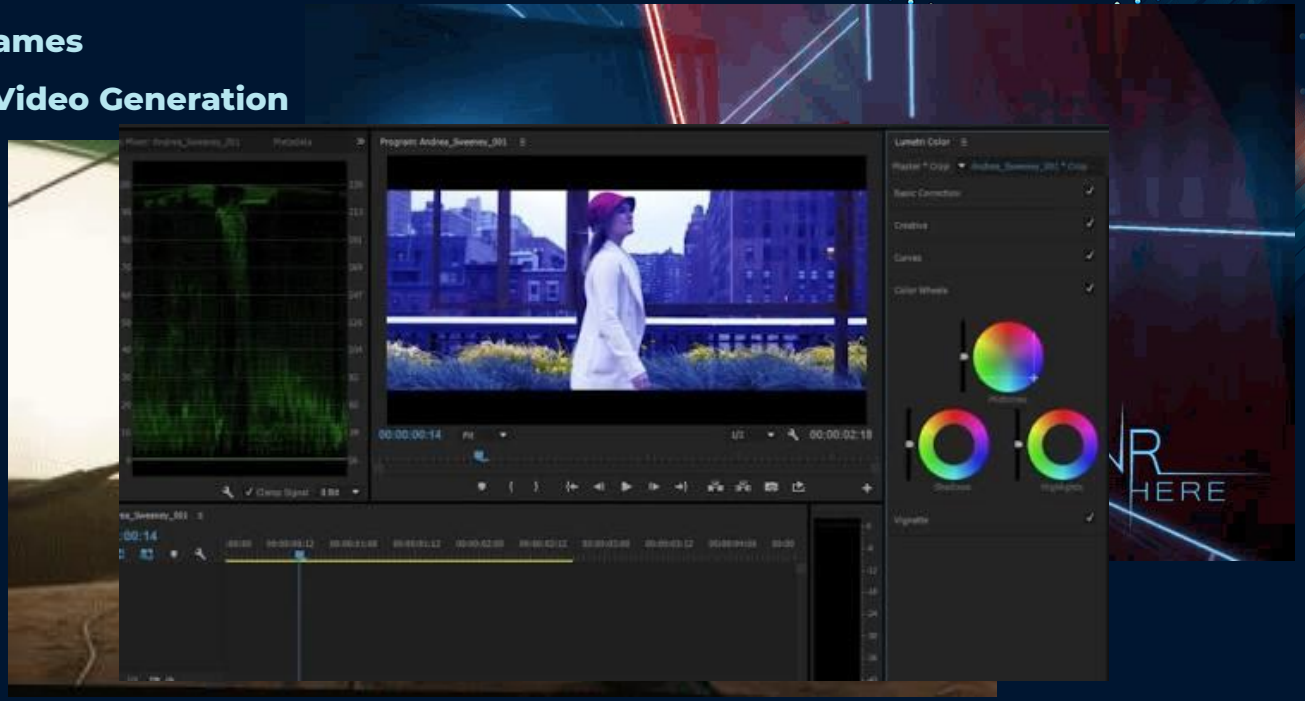




Application

Arts, Media & Entertainment

- Special Effects in Film and Animation
- Augmented Reality Games
- Automated Image or Video Generation



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Tasks of Computer Vision

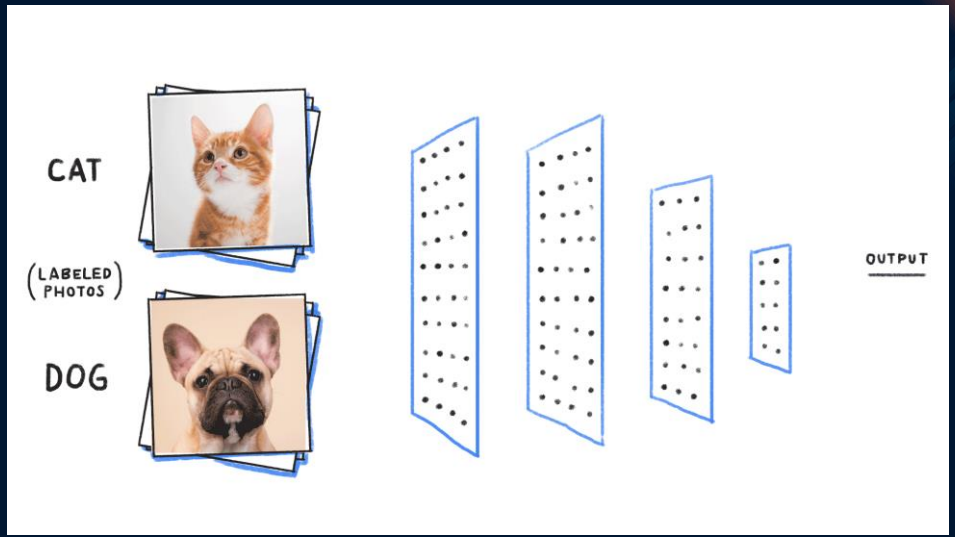


Tasks of Computer Vision

1.Image Classification

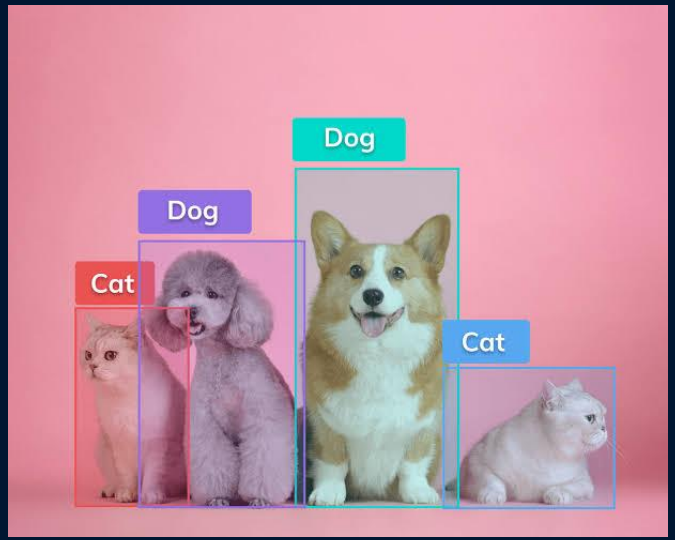


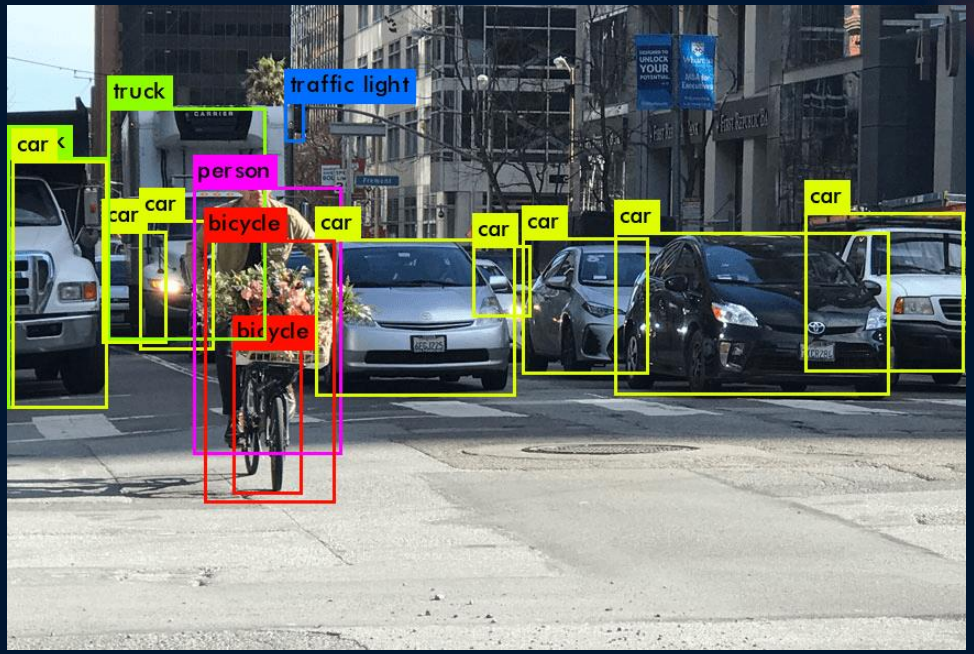
Assign a single label to the entire image



2.Object Detection

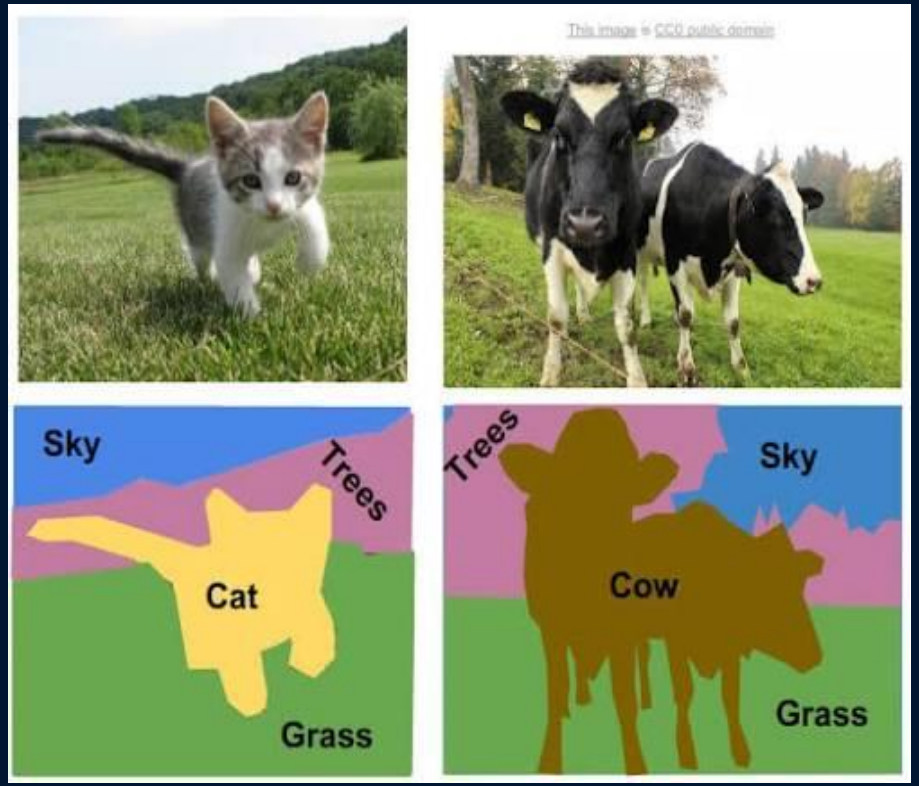
Object detection = Label + Bounding Box





Tasks of Computer Vision

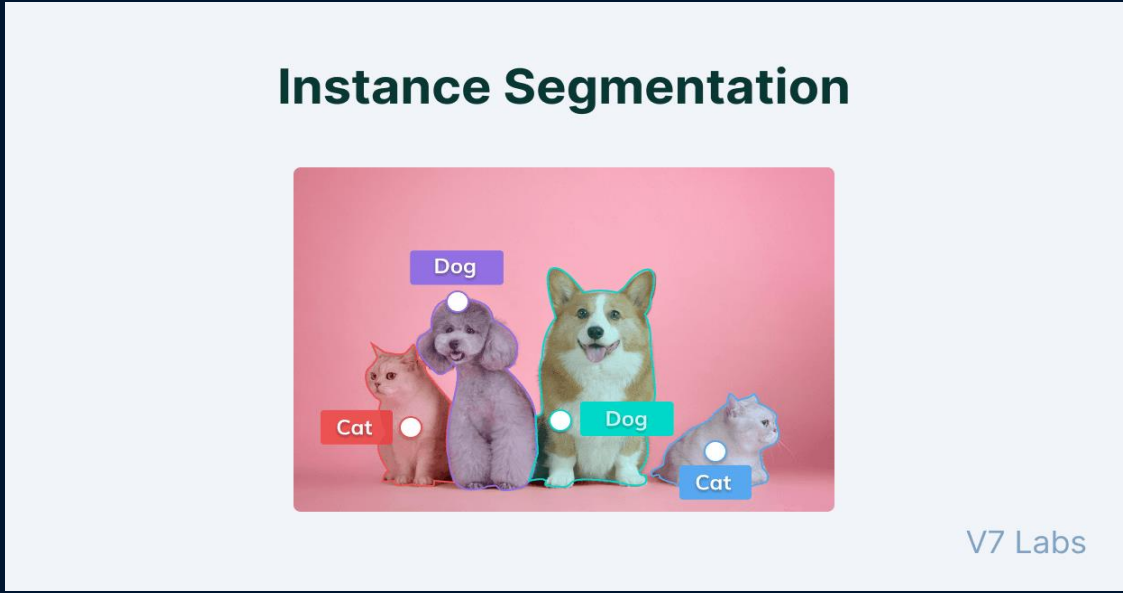
3.Semantic Segmentation



Assign a class label
to each pixel
in the image
Example:
Sky
Cat
Trees
Grass

4.Instance Segmentation

Detect and segment individual object instances
Example: Two cats and two dogs are shown separately



Classification



CAT

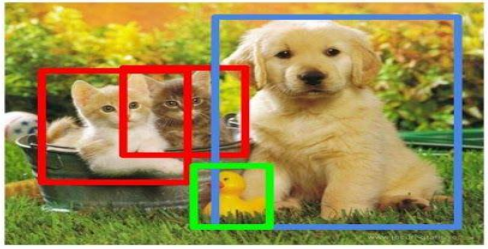
Classification + Localization



CAT

Single object

Object Detection



CAT, DOG, DUCK

Instance Segmentation



CAT, DOG, DUCK

Multiple objects

Semantic Segmentation



GRASS, CAT, TREE, SKY

No objects, just pixels

Classification + Localization



CAT

Single Object

Object Detection



DOG, DOG, CAT

Instance Segmentation



DOG, DOG, CAT

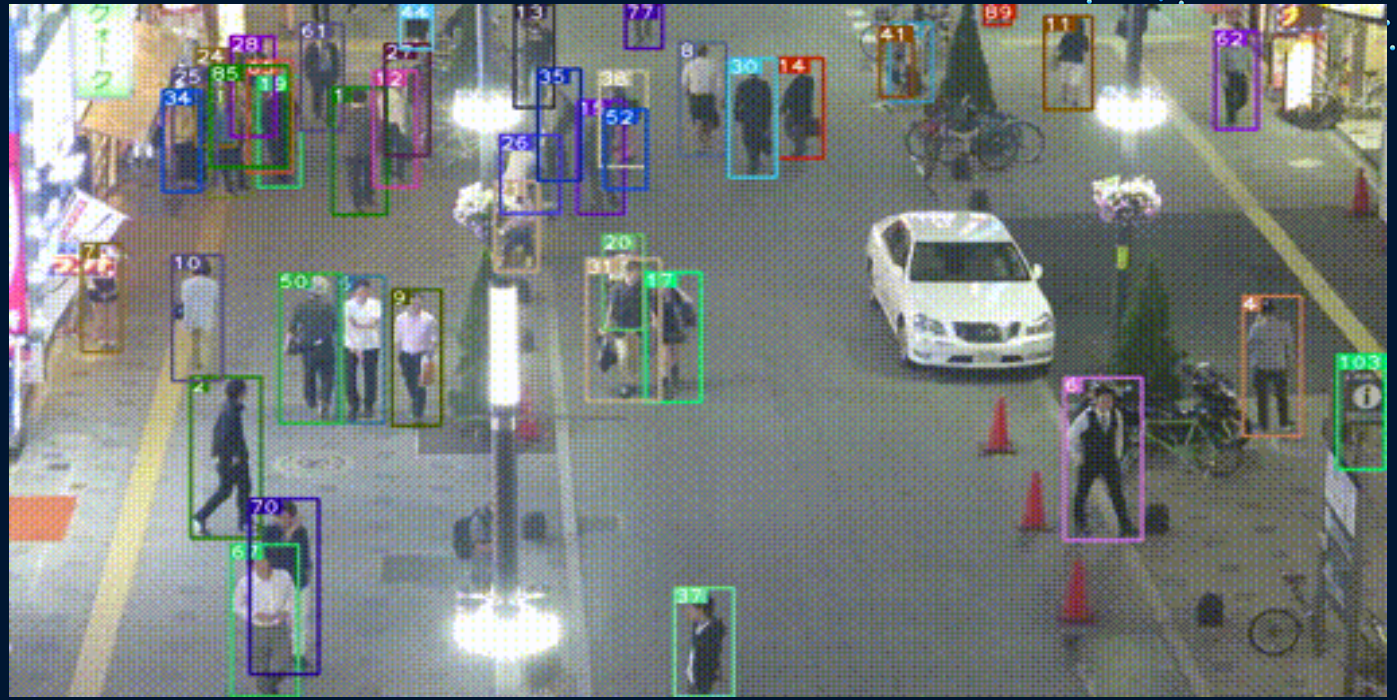
Multiple Object

This image is CC0 public domain

Tasks of Computer Vision

5.Object Tracking

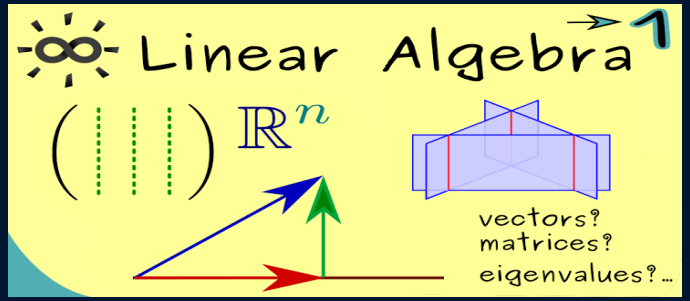
Follow an object across frames in a video

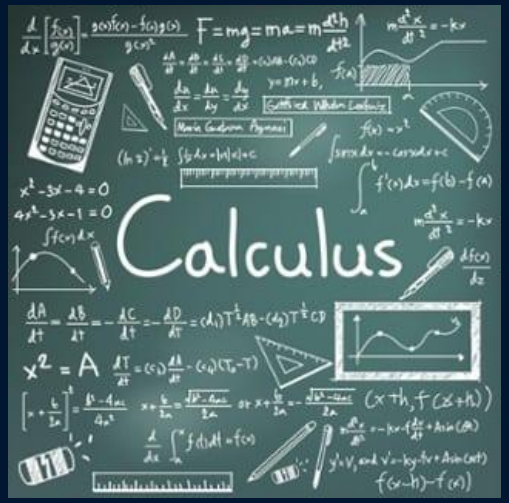


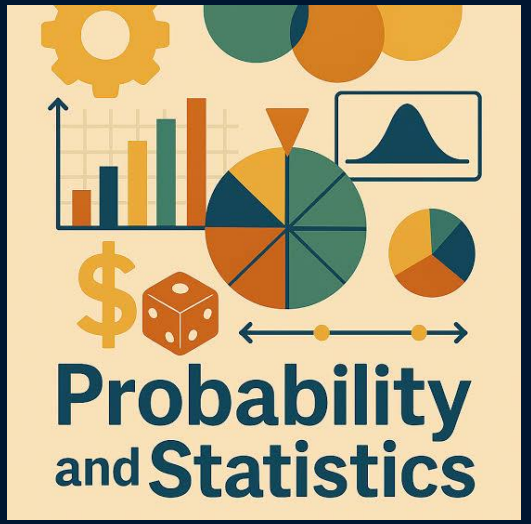
Roadmap



1. Mathematics





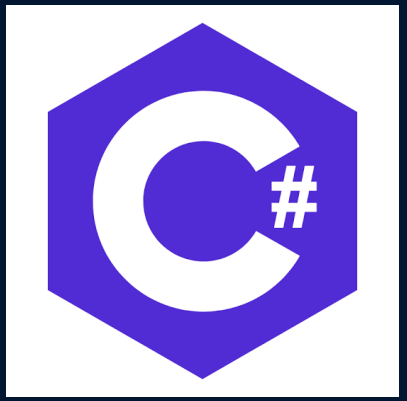


2. Programming









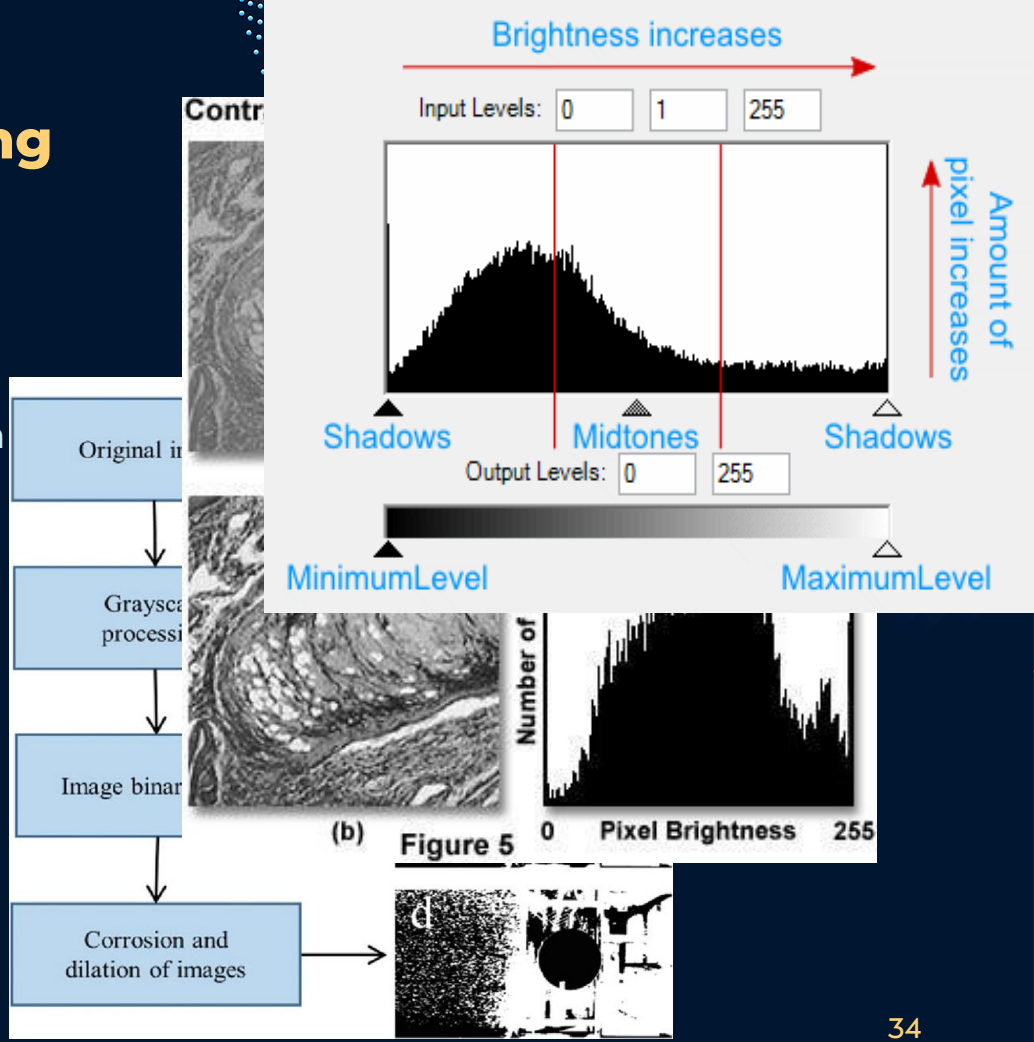




Roadmap

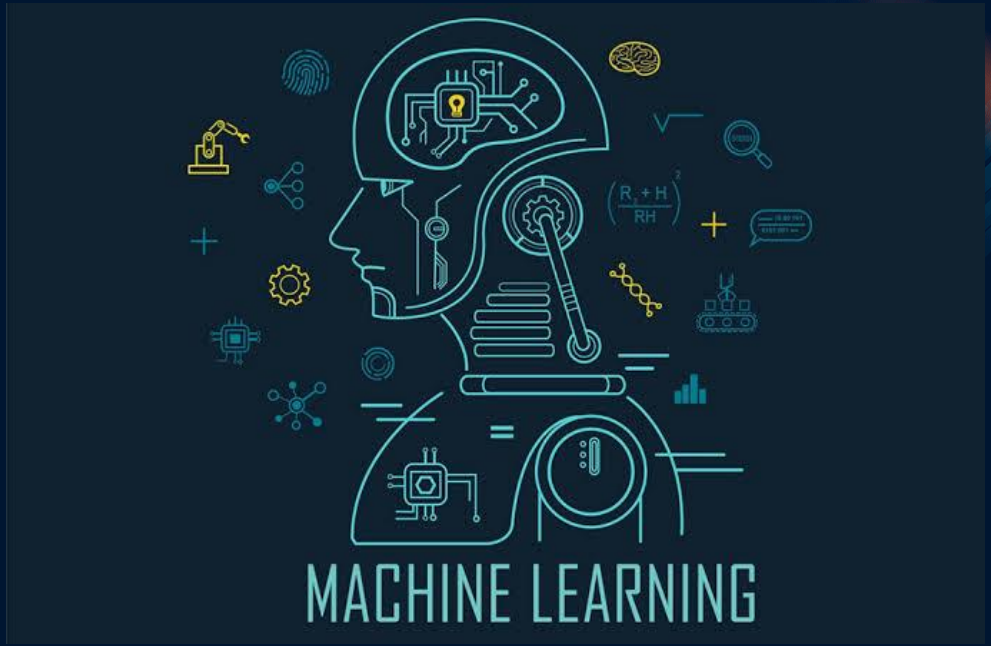
3. Classical Image Processing

- Digital Image Basics
- Preprocessing
- Resize, noise reduction, normalization
- Filtering & Edges
- Convolution, edge detection
- **Transforms:** Fourier, Hough
- **Features:** SIFT, SURF, ORB, HOG



4. Machine Learning

- Basic concepts of machine learning
- Machine learning algorithms
- Data Preparation
- Feature Engineering
- Evaluation criteria and metrics



Roadmap

5. Deep Learning

- Basic concepts of neural networks
- Convolutional Neural Networks (CNNs)
- Regularization and improvement techniques
- ResNet, VGG, RNN, ...
- Transfer Learning
- Fine-tuning
- Computer vision architectures



Tools and Resources



1. Libraries & Frameworks











TensorFlow

2. Platforms & Online Tools

kaggle



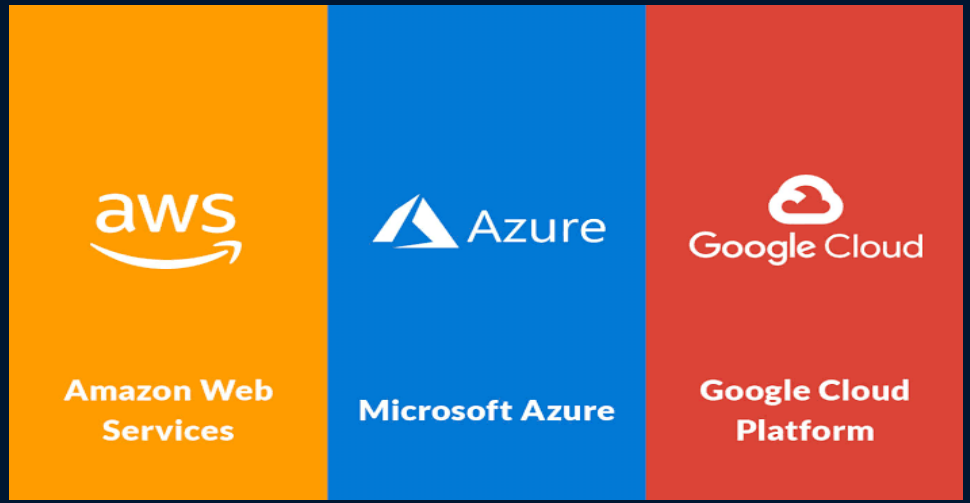


Tools and Resources

3. Cloud / Execution

Google colab

kaggle NOTEBOOKS



4. Learning Websites







Thank You

About me:
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Computer vision developer



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